ExplainableCropAI: Multi-level Explanation Process

From Input Image to Actionable Agricultural Decision Support

ExplainableCropAI Components
- Input/Output
- Model Component
- Visual Explanation
- Uncertainty Quantification
- Natural Language Explanation
- Recommendation Engine
- Key Information

Input Image

Paddy leaf image
captured in the field

Image input

Disease Detection Model

Modified ResNet50
Prediction: Brown Spot
Confidence: 92%

Feature maps

Multiple forward passes

Visual Explanation

Visual transparency for verification

Grad-CAM highlighting
important regions
contributing to prediction

Uncertainty Quantification

Honest uncertainty for appropriate trust

Monte Carlo Dropout
Epistemic: Low
Aleatoric: Medium
Entropy: 0.32

Visual evidence

Confidence level

Natural Language Explanation

This image shows a paddy plant affected by
Brown Spot
disease, as indicated by the
circular brown lesions with yellow halos
visible on the leaf surface. The system is
moderately confident (92%)
in this diagnosis because the distinctive pattern of spots. Brown Spot typically develops during periods of
high humidity
and can be distinguished from Bacterial Leaf Streak by its circular rather than angular lesions.

Diagnosis + Uncertainty

Context-Aware Recommendation

TREATMENT RECOMMENDATION:

1. Apply fungicide containing
propiconazole
(e.g., Tilt) at 1ml/L water.
2. Maintain adequate field drainage
3. Monitor adjacent plants for early symptoms.

Explanation text

Treatment options

Decision Support Output

Domain-specific language for accessibility

Context-aware advice for practical utility

INTEGRATED ANALYSIS & RECOMMENDATION

Disease:
Brown Spot
(92% confidence)
Visual Evidence:
Circular lesions
with yellow halos
Severity:
Moderate
(30% leaf area affected)
Recommendation: Apply
propiconazole
fungicide
Uncertainty Note:
Medium uncertainty
in severity assessment; consider expert consultation if symptoms worsen

SEVERITY: Moderate
(30% leaf area affected)

ALTERNATIVE:
For organic farming, apply neem oil solution at 2-3% concentration.

to reduce humidity.

Integrated support for decision-making

Actionable insights

